

OPEN SOURCE MONITORING CONFERENCE | NOV 18–20 2025

ONOTIFY: A SCALABLE, FLEXIBLE ALERTMANAGER

COLIN DOUCH
DUCKDUCKGO



Colin Douch • 2:25 PM

Thanks! It's gonna be a _lot_ of work modernising their Observability stuff - they still use Graphite and Nagios!

[redacted]

It's so old school that I don't know what those are 😂

Which brings me to *Alertmanager*

- Alertmanager *infuriates me*
- What happened to all my nice alerting features?
- Why am I being forced to use this thing?



DuckDuckGo

Alertmanager is this weird little orphan child

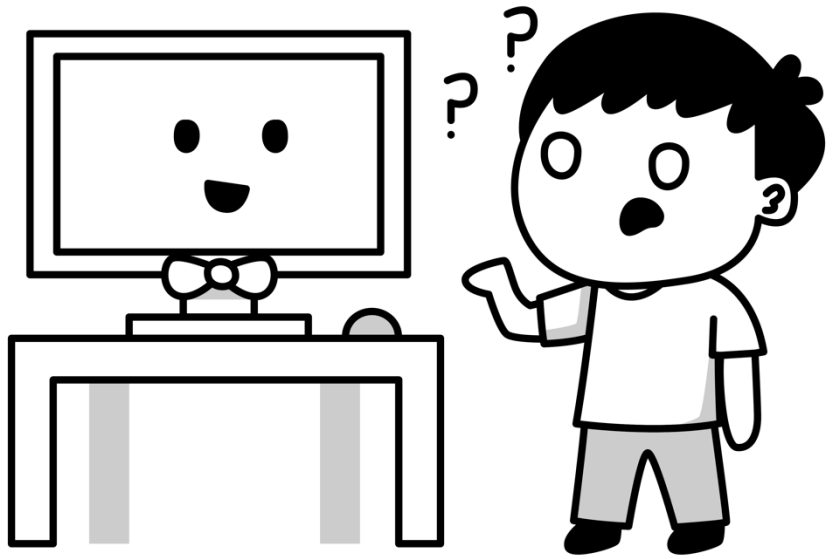


- It’s had a *fifth* of the commits of Prometheus itself
- People are sort of forced to use it if they use Prometheus
- Providers are bullied into helping maintain it because of Prometheus’ success

DuckDuckGo

So, let’s talk about it

- Alertmanager’s weird incentives
- What we’ve lost
- Why is Alertmanager so difficult?
- Can we do better?

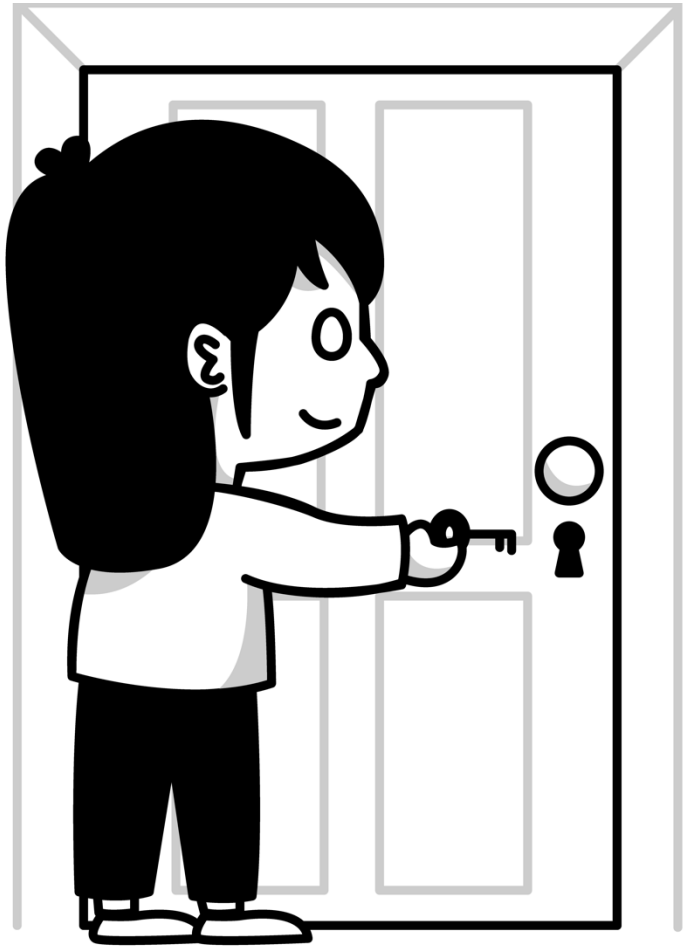


Hi! I’m Colin



- Formerly leading the Observability Team at Cloudflare
- Now an SRE at DuckDuckGo, working on privacy preserving telemetry at scale
- Older than I look
- My knees hurt

What about Authentication?

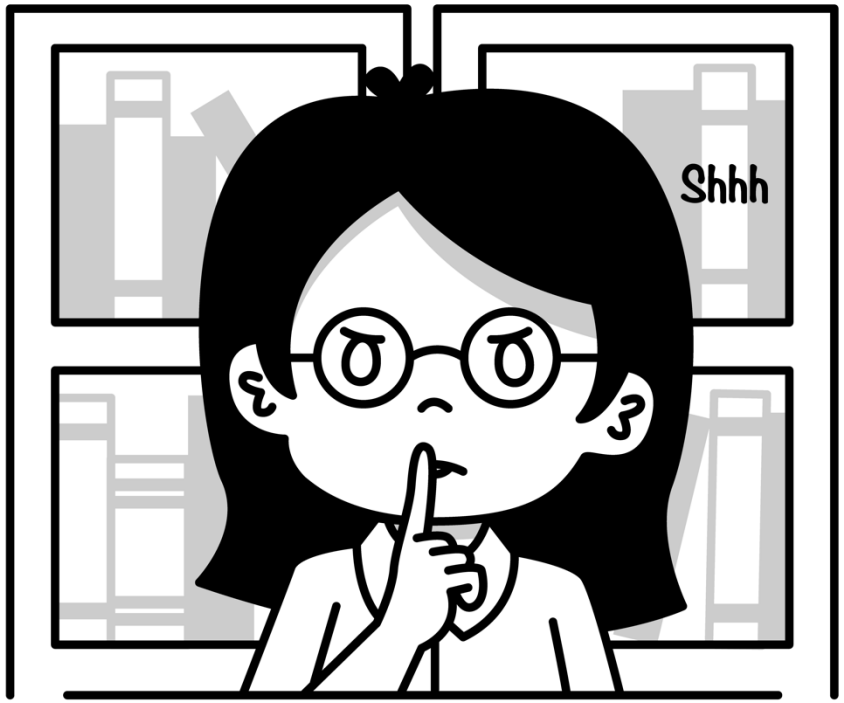


- Believe it or not, it’s useful to know who does things
- Alertmanager, when it can sort of relies on an honor system
- But for the most part, you’re on your own. Who made changes? Who knows! Not you!

DuckDuckGo

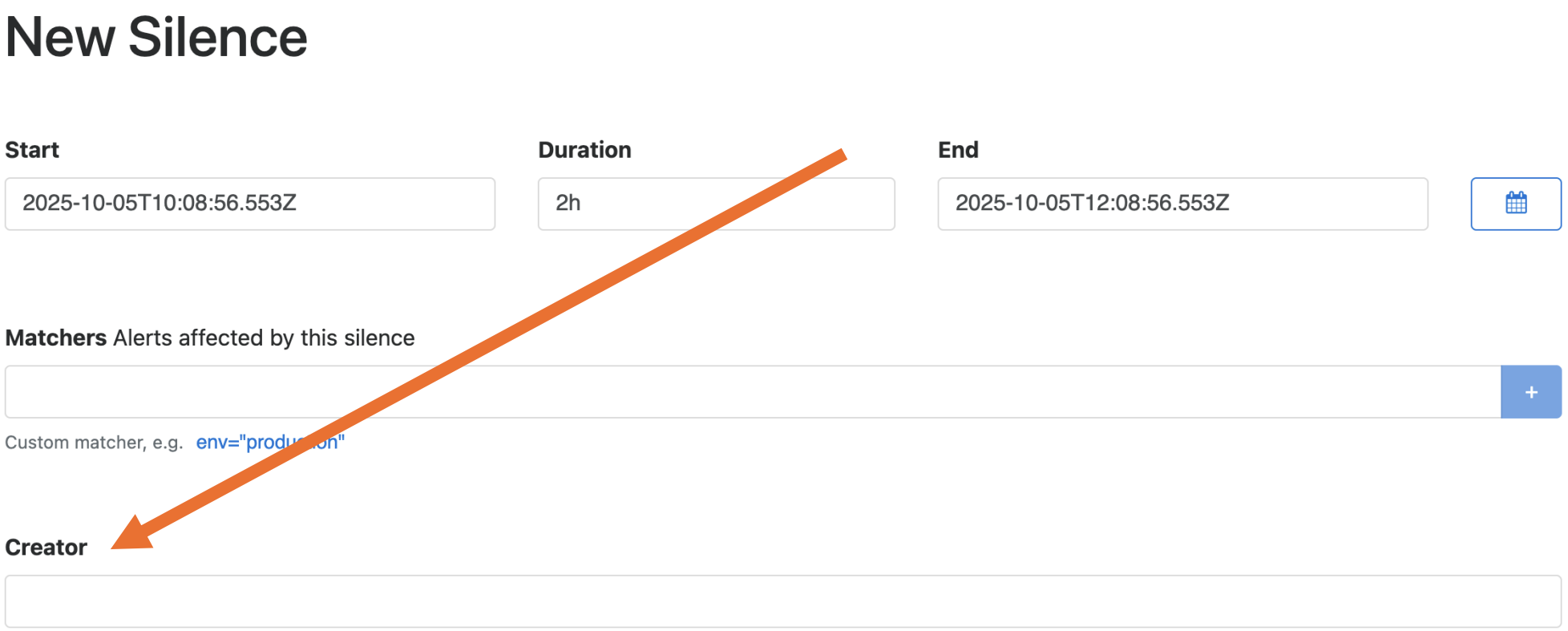
What about Silences? Or downtimes?

- Maybe you’re doing some maintenance
- Maybe you’re getting a PTSD response to your pagerduty alert noise

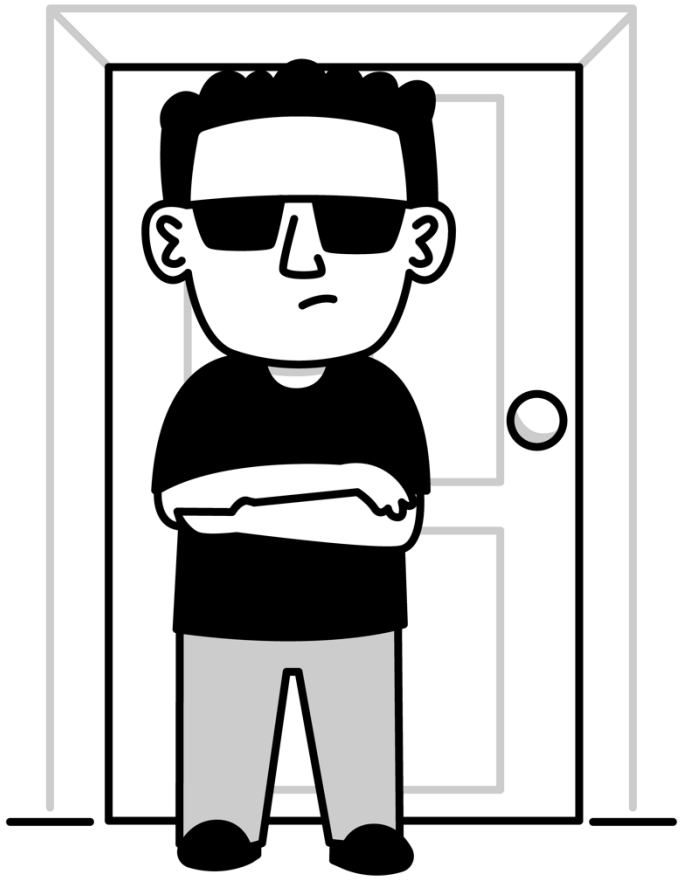


DuckDuckGo

What about Silences? Or downtimes?



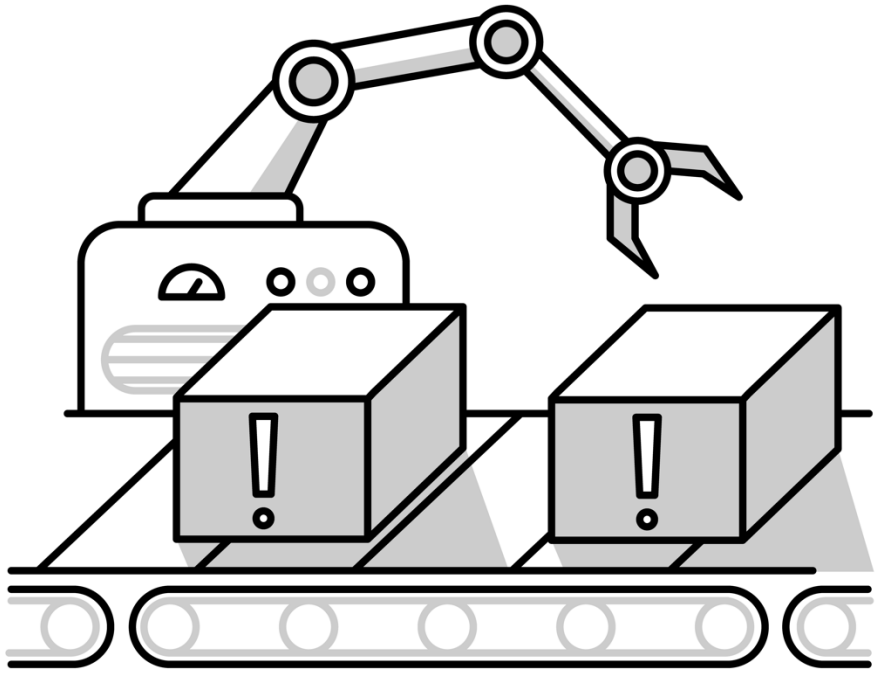
“If you like it you shoulda put a proxy in front of it”



- Alertmanager recommends putting a proxy in front of it for any business logic
- But it notably doesn’t actually provide any such proxy
- We made ”Alertmanager Bouncer” at Cloudflare: https://github.com/sinkingpoint/alertmanager-bouncer

Automation? We had Event Handlers for that

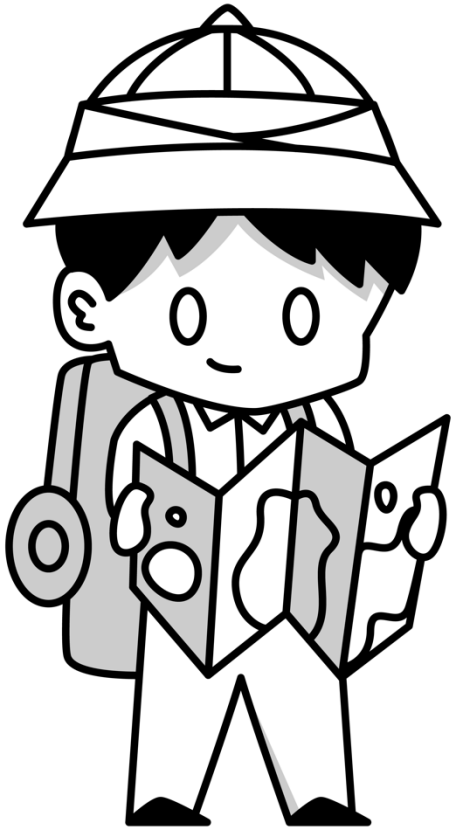
- Automating Alert Remediation (***temporarily*** while a real fix is being implemented) is sort of a core DevOps tenant
- We're left on our own, yet again with Alertmanager
- Better get used to the Alertmanager Webhook interface
- Good luck



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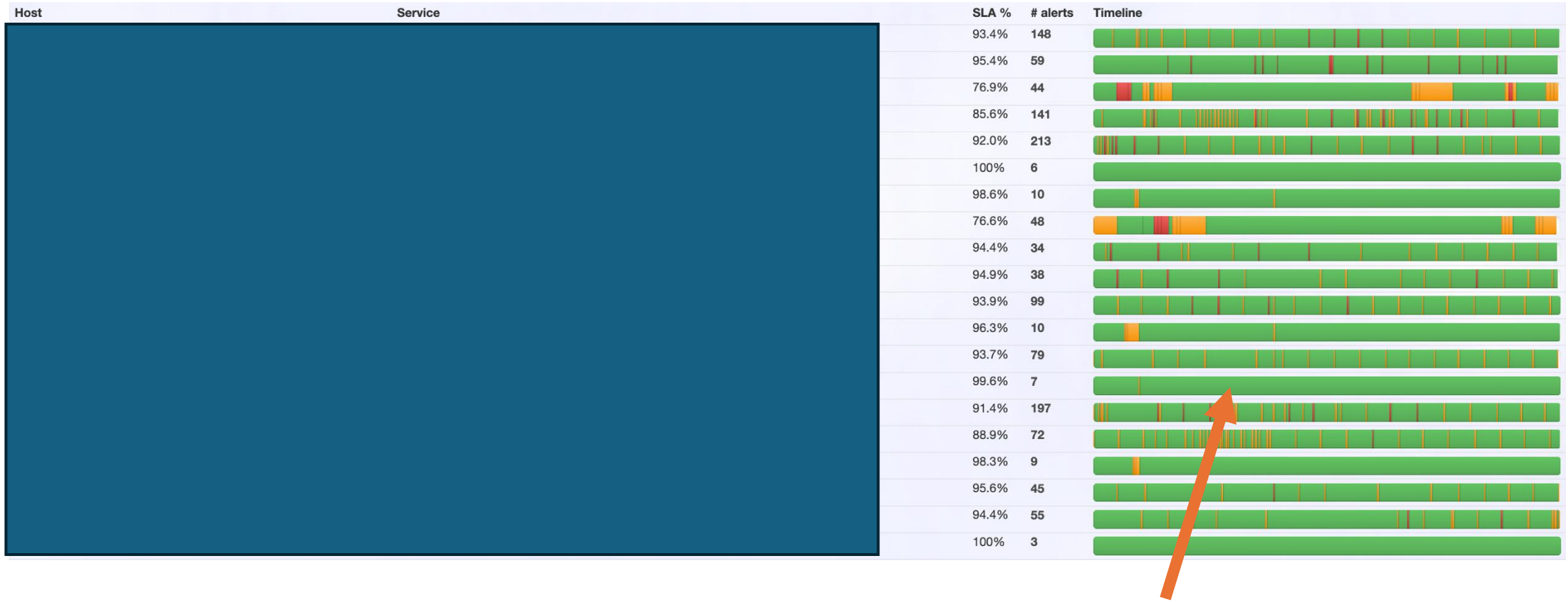
What about Alert Histories?

- Yet another hard coupling with Prometheus – you can just use the ALERTS timeseries
- But what if you have multiple Prometheus servers?
- What if you need longer retentions?



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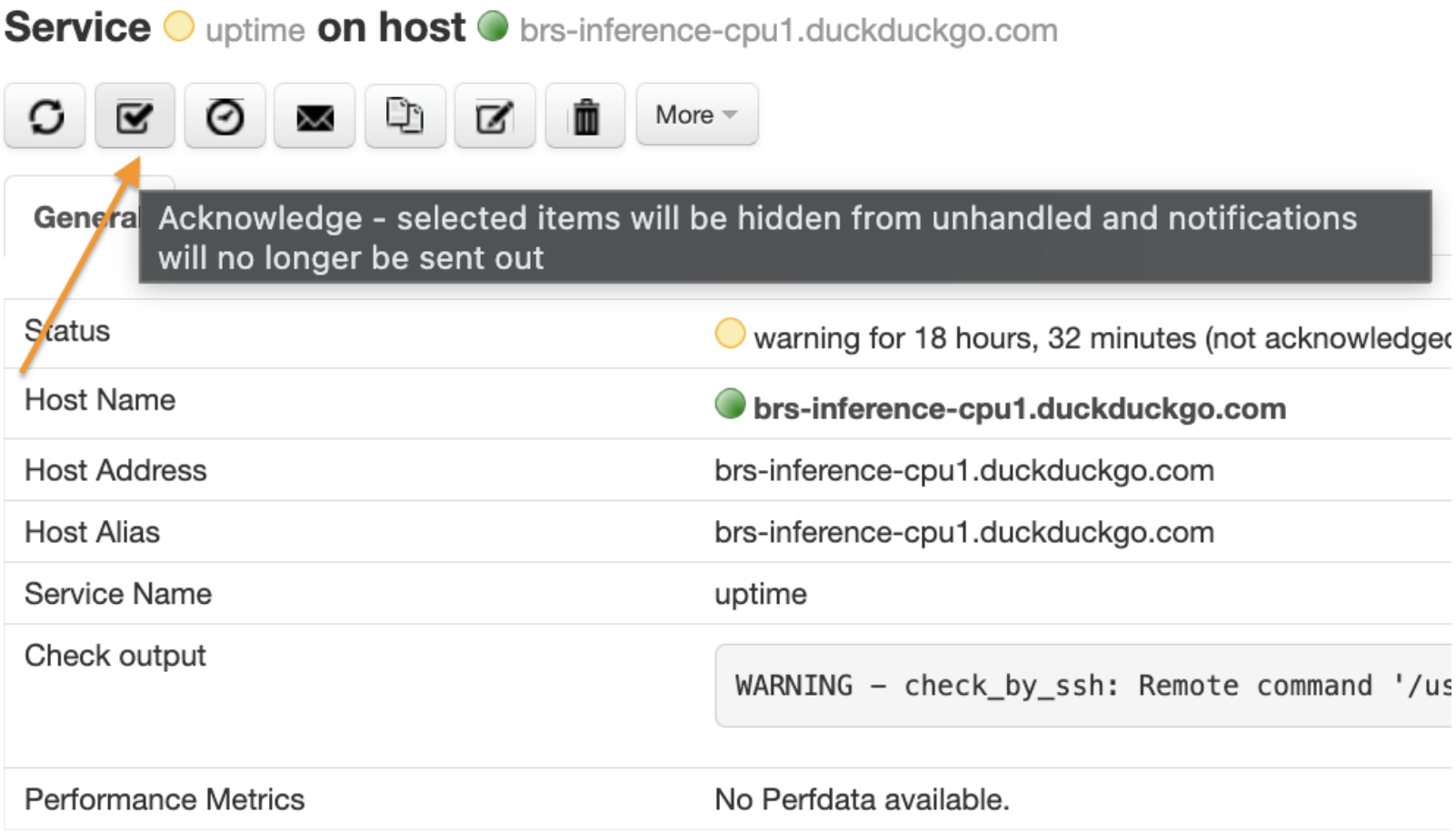
What about Alert Histories?



Histories!!!

DuckDuckGo

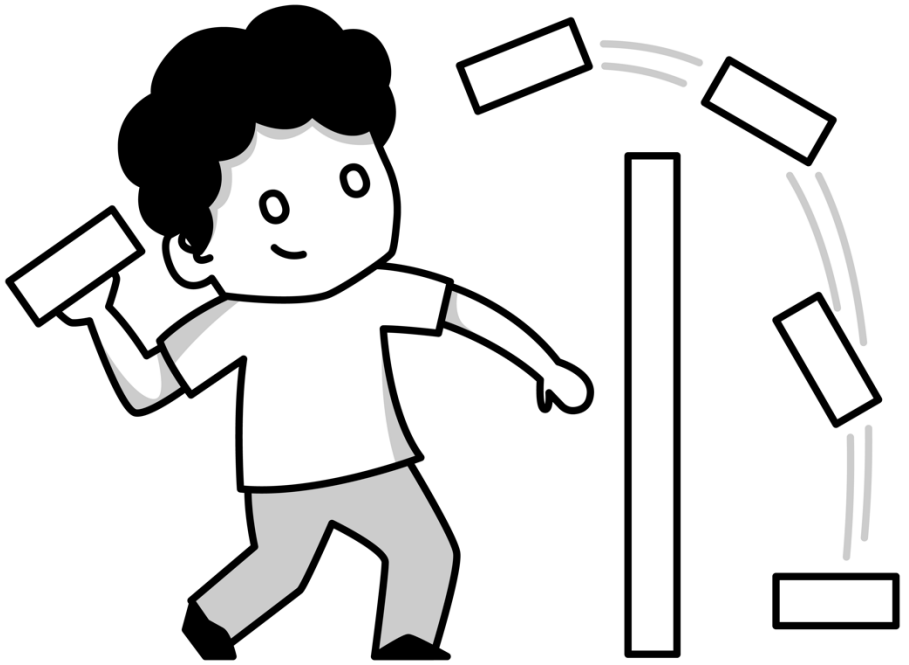
Let’s talk about Acknowledging



DuckDuckGo

Let’s talk about Acknowledging

- Once again, Alertmanager punts this to the downstream sinks
- Fine, until you have multiple alerting systems
- Oh look, yet more vendor lock in



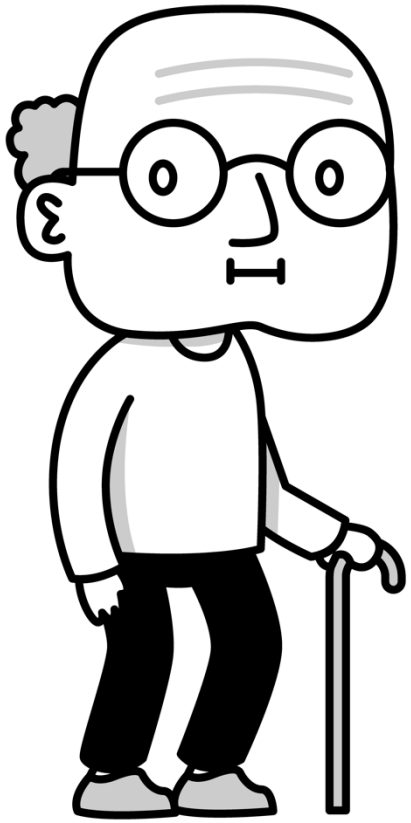
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Which brings us back to: AlertManager is *weird*

- Which brings us back to this: Alertmanager is *weird*
- It's almost *toxically* open source
- Providers are both incentivized to help maintain Alertmanager, and simultaneously do *as little as possible* to it



We've lost *so much*



- All our functionality is offloaded to downstream alerting sinks
- Providers are incentivized to put as little work into Alertmanager as necessary
- This is great for vendor lock-in, and bad for us who actually have to *use* Alertmanager

Which brings me to onotify

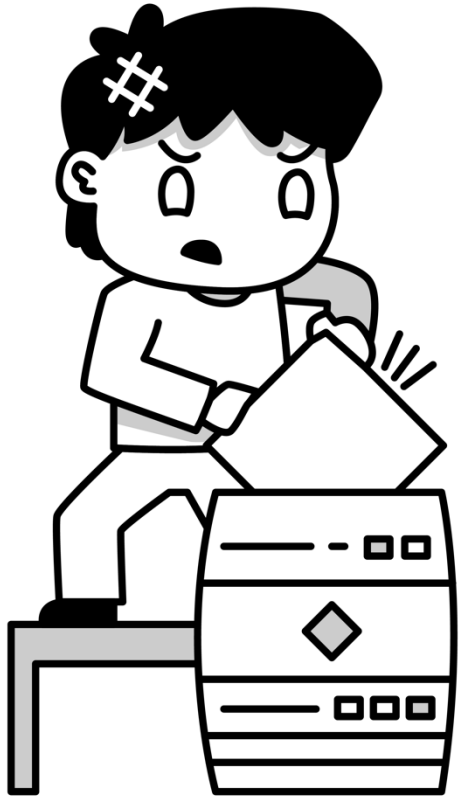


https://github.com/sinkingpoint/onotify

DuckDuckGo

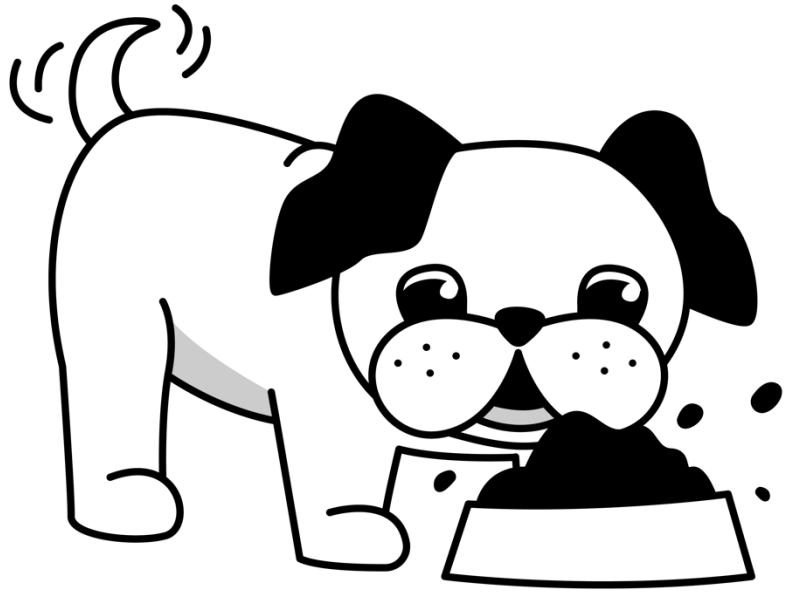
What we've done differently

- Kiora attempted to mimic Alertmanager as much as possible
- Onotify acknowledges:
  - Alerting is *bursty*
  - Alertmanager is *wasteful*
  - There's a lot of effort already put into Alertmanager both in Open Source, and internally



DuckDuckGo

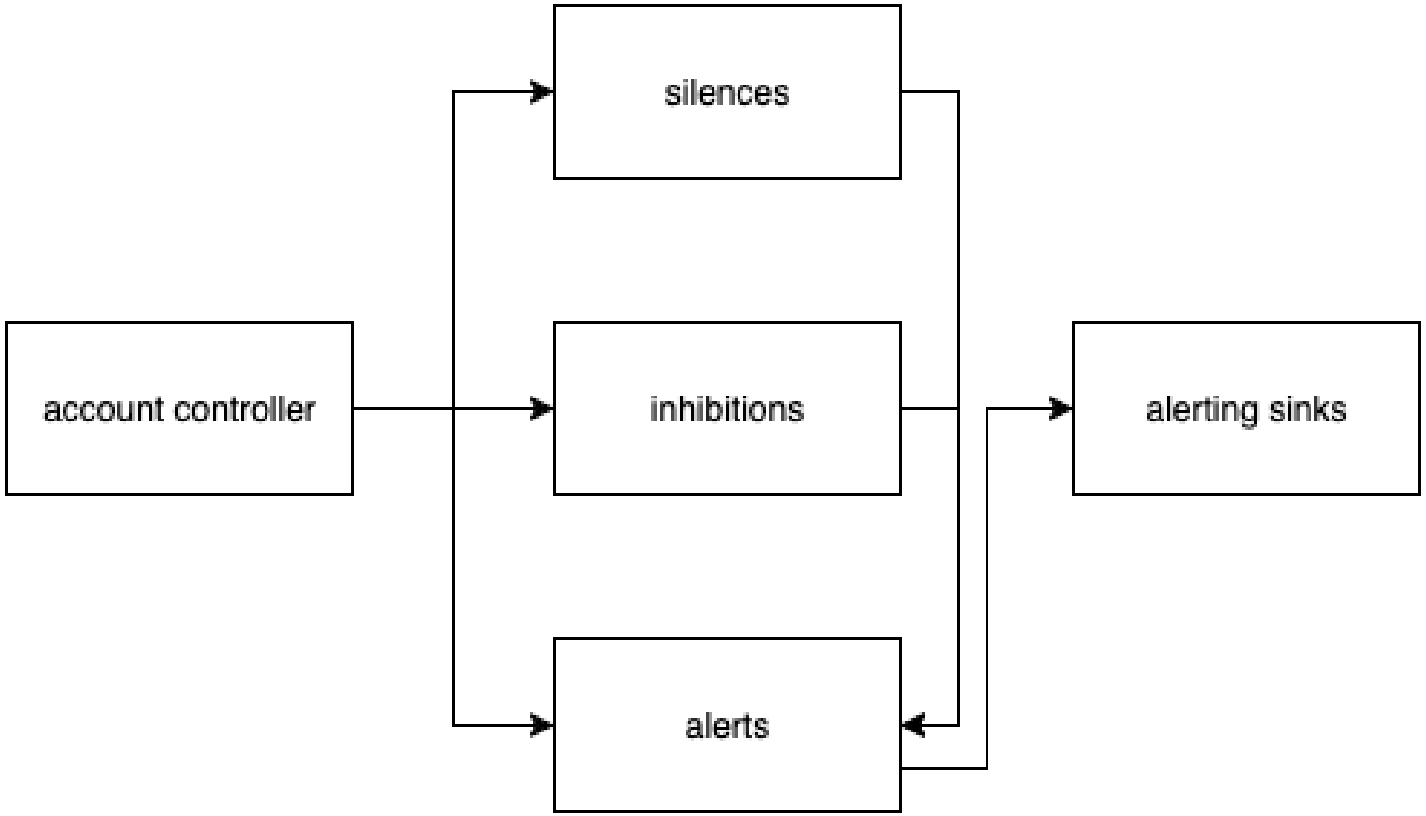
Serverless to the rescue!



- Have bursty traffic? Serverless might be for you!
- Cloudflare is big on dogfooding, so Cloudflare Workers it is
- Not just a serverless platform: Lots of nice primitives to build full applications, like Durable Objects

DuckDuckGo

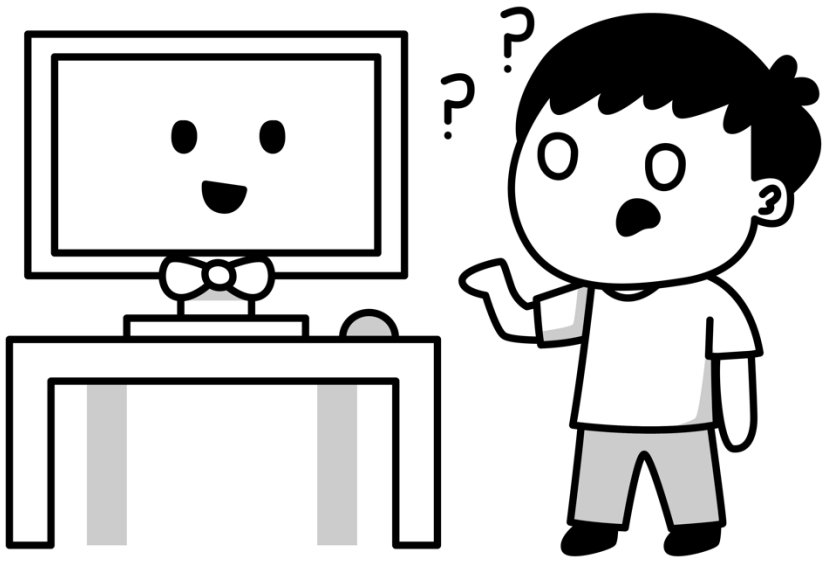
Onotify has logical separation by *“account”*



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What about all the existing stuff?

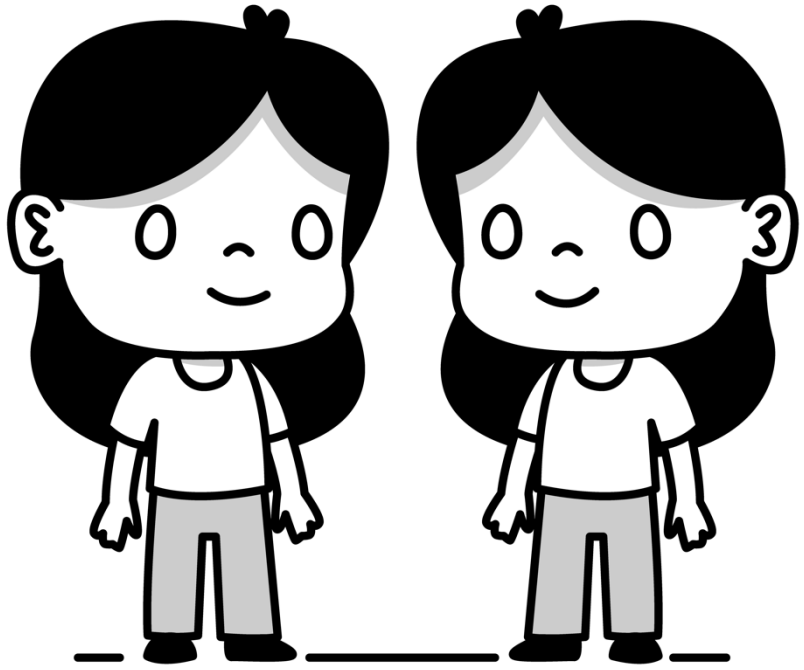
- There's a lot of Alertmanager stuff out there!
- Despite its flaws, there *is* a large community effort to make up for it
- Karma, amtool, Alertmanager Bouncer, all your existing config, there's lots of effort put into them!



DuckDuckGo

onotify needed to mimic Alertmanager

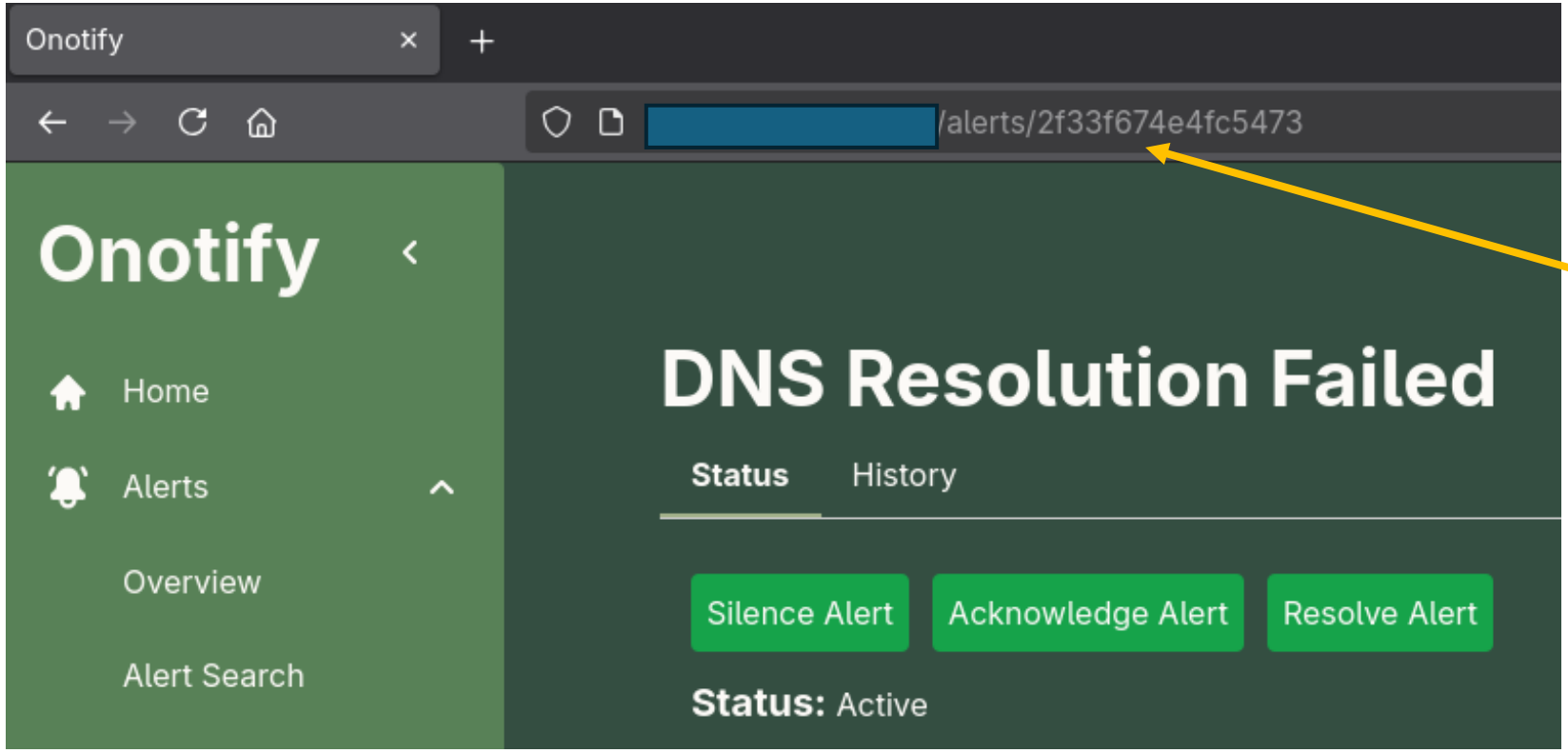
- onotify needed to look as much like Alertmanager as possible
- It takes the same config
- It’s entirely backwards compatible with Alertmanagers API
- Karma, amtool, your random script that keeps everything running? It “just works”



DuckDuckGo

But we still improved things

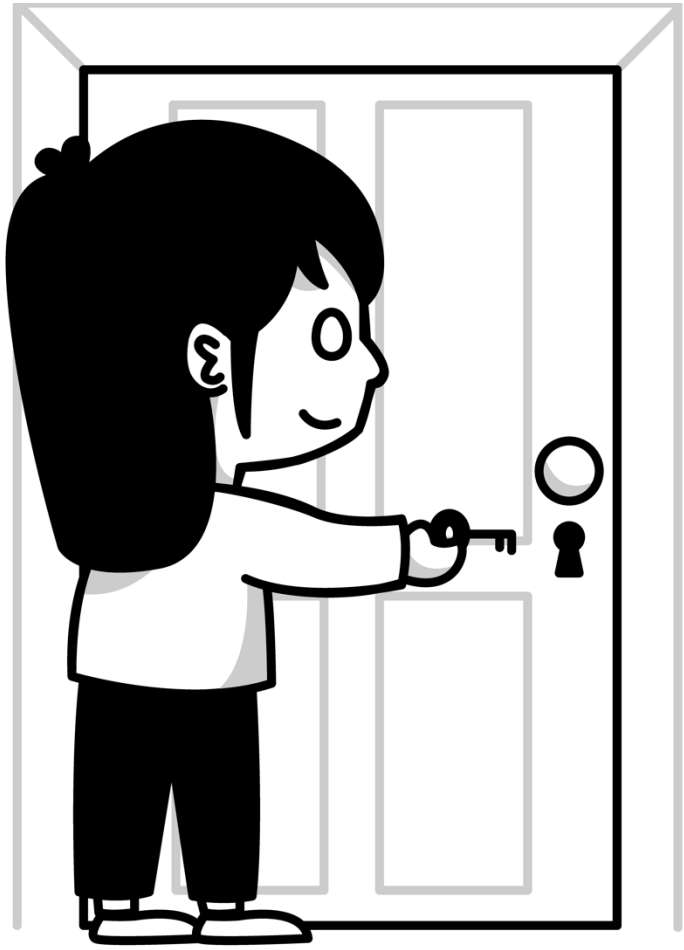
We have *links*!



Canonical Links

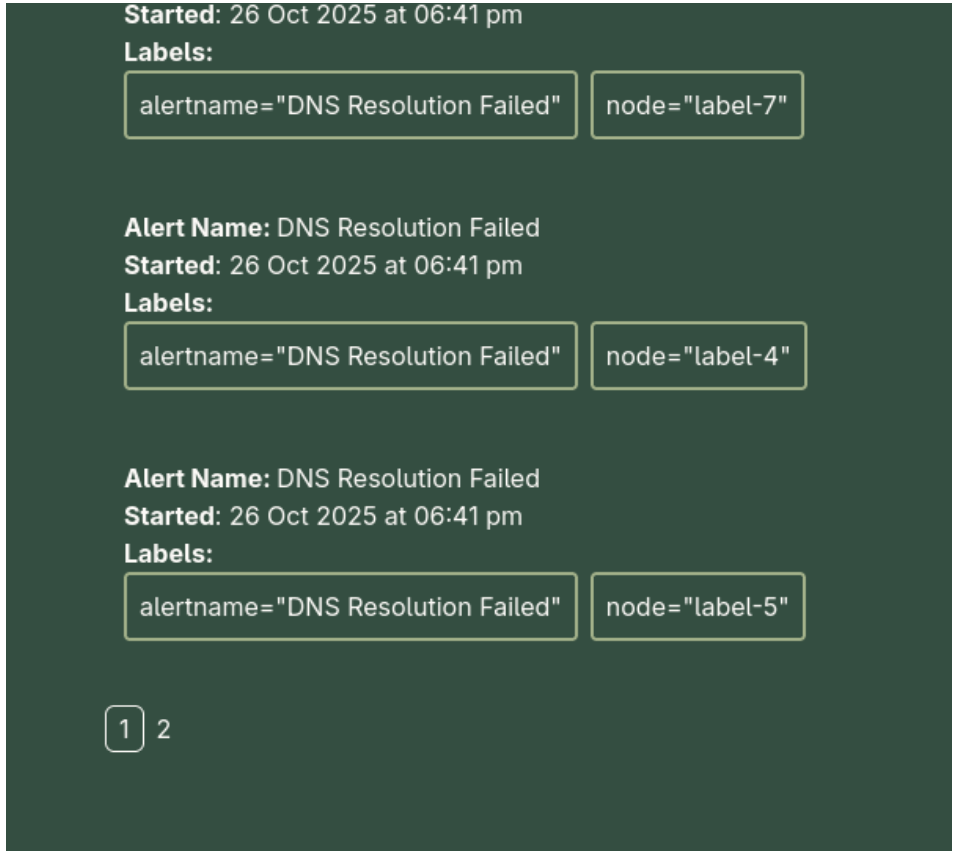
We have authentication!

- onotify authenticates every request - no more giving everything access to everything
- It supports its own authentication, or farming it out to Cloudflare Access or Clerk
- No more honor system!



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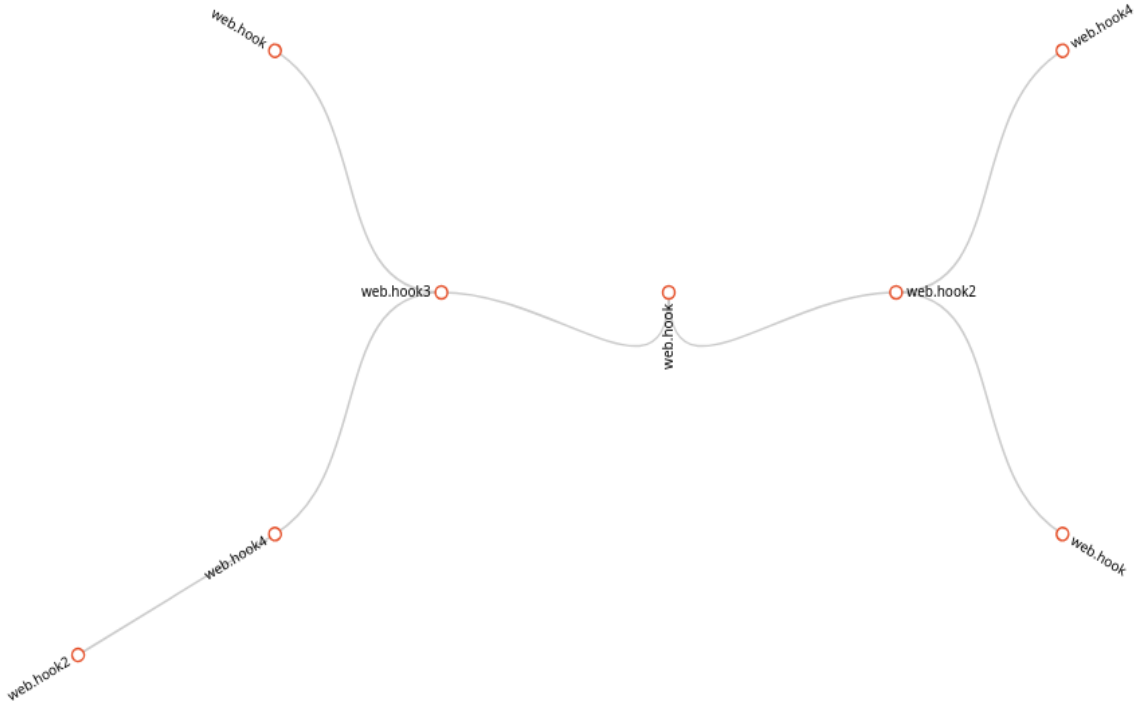
Requests have pagination



- onotify is built to scale!
- Pagination
- Authentication/authorization
- Centralized Alert histories, comments, and acknowledgements

What about configuration?

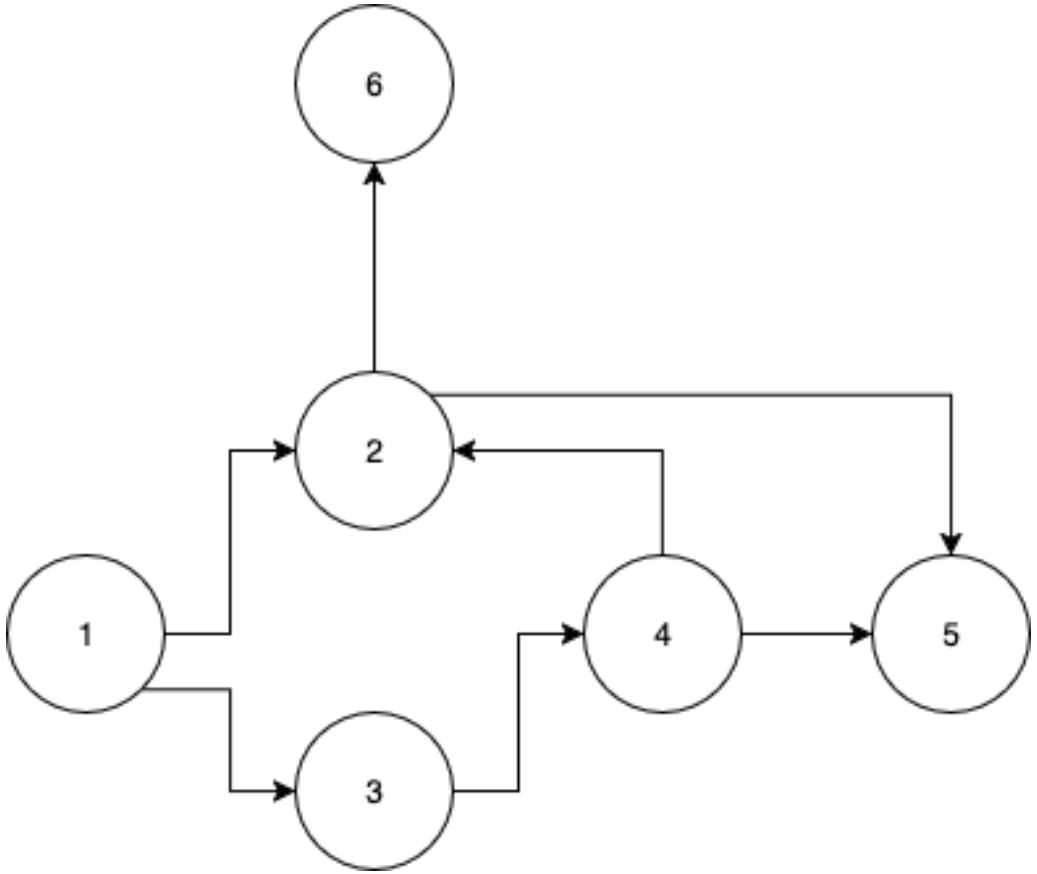
- You can upload your Alertmanager configuration directly
- But at Cloudflare, we had *lots* of duplicated branches
- Our alert evaluations started to push *10 seconds*



DuckDuckGo

What about configuration?

- But no more routing trees!
- Instead: The routing Directed Acyclic Graph (DAG)
- Faster to evaluate, no duplication, less resource usage, easier updates

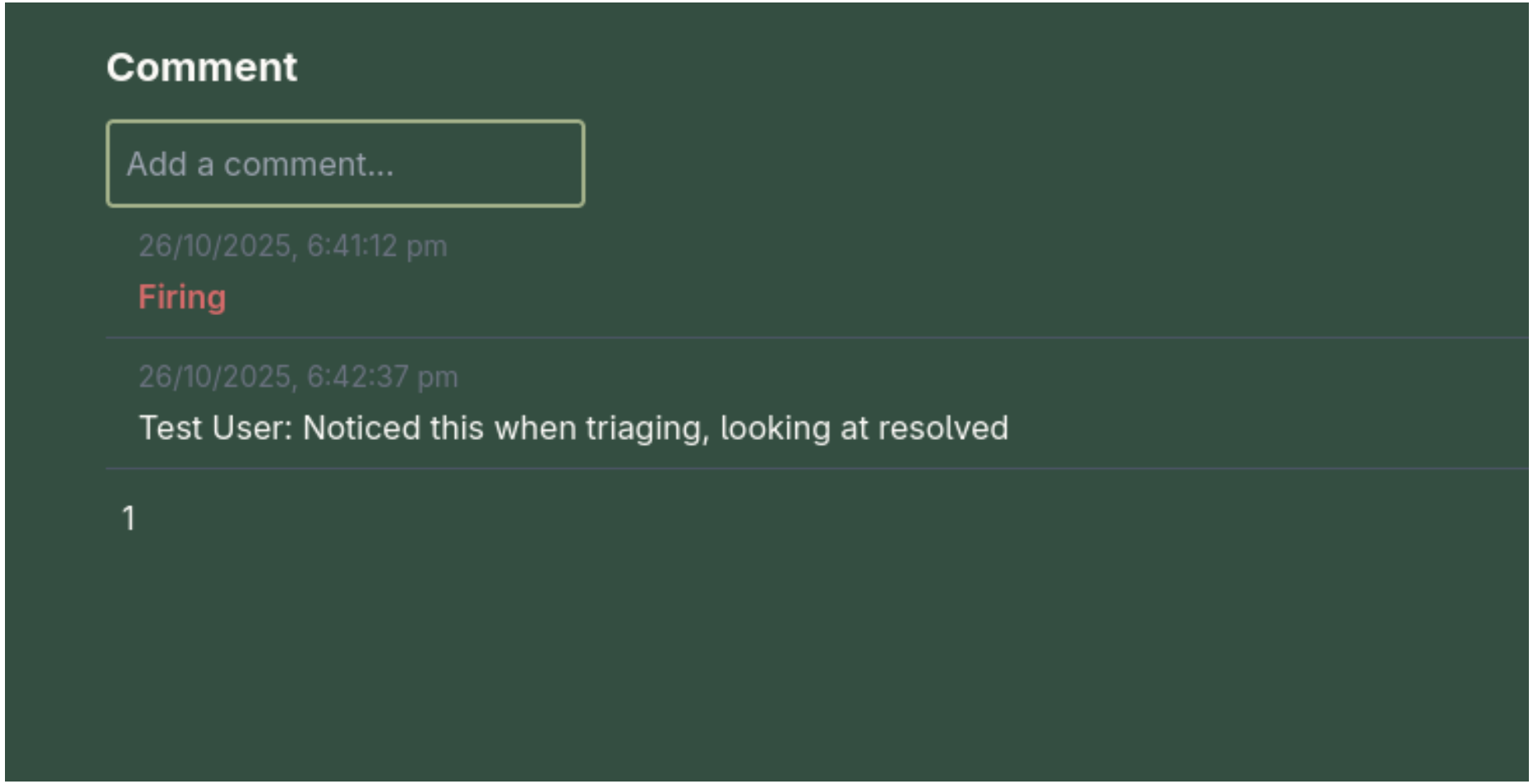


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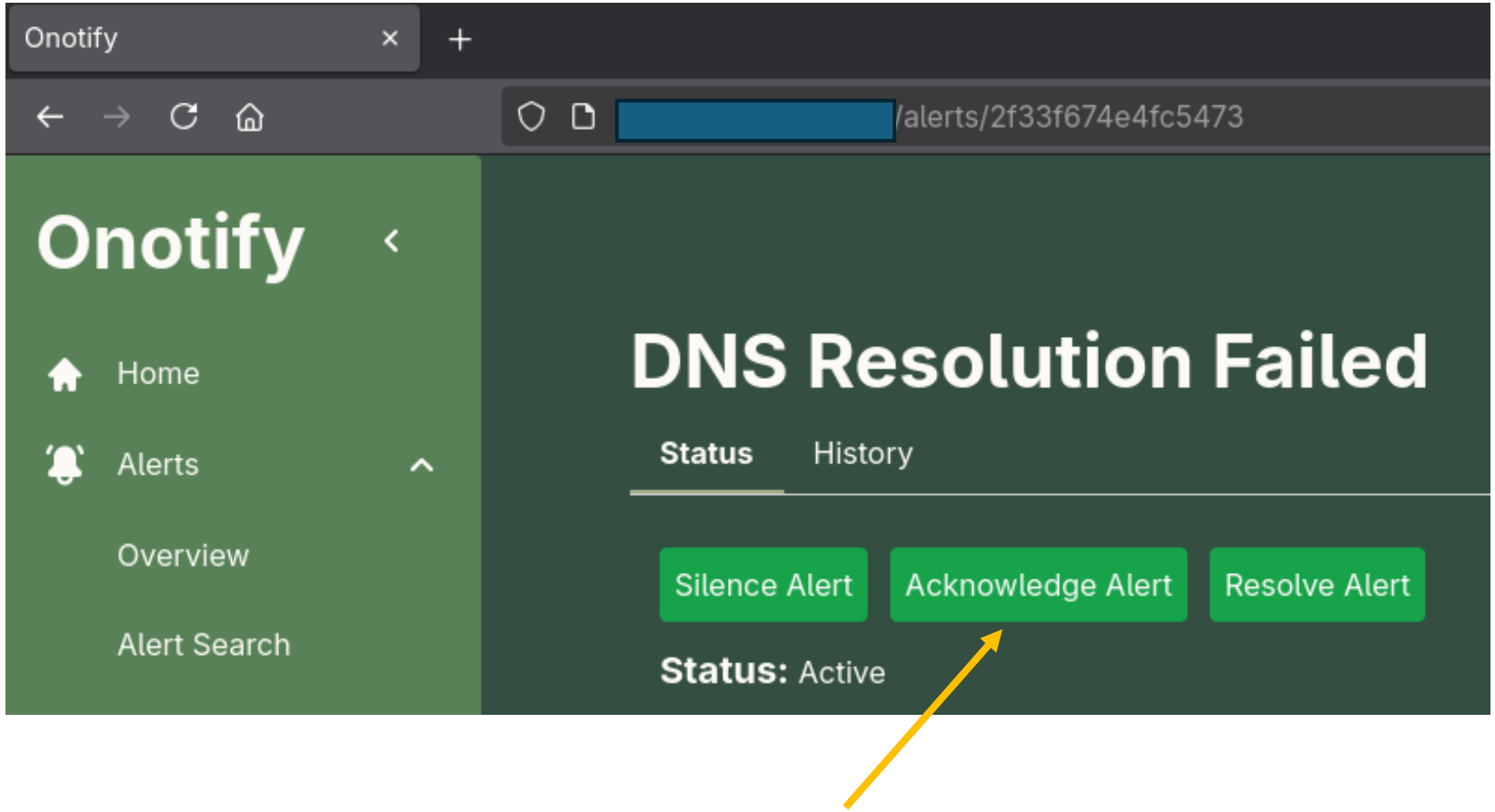
onotify has proper histories!



We can annotate alerts



And we can acknowledge them too!



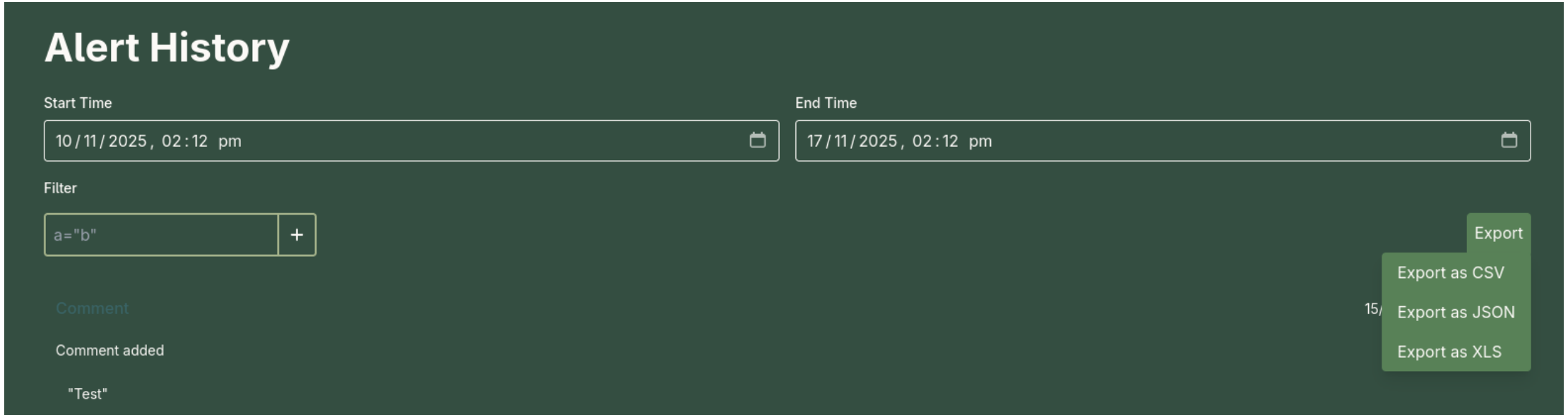
onotify centralizes acknowledgements

- You can acknowledge alerts in onotify
- Those acknowledgements can be viewed in the history
- They can be *synced* with downstream providers – no more checking multiple places



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We can even export them!



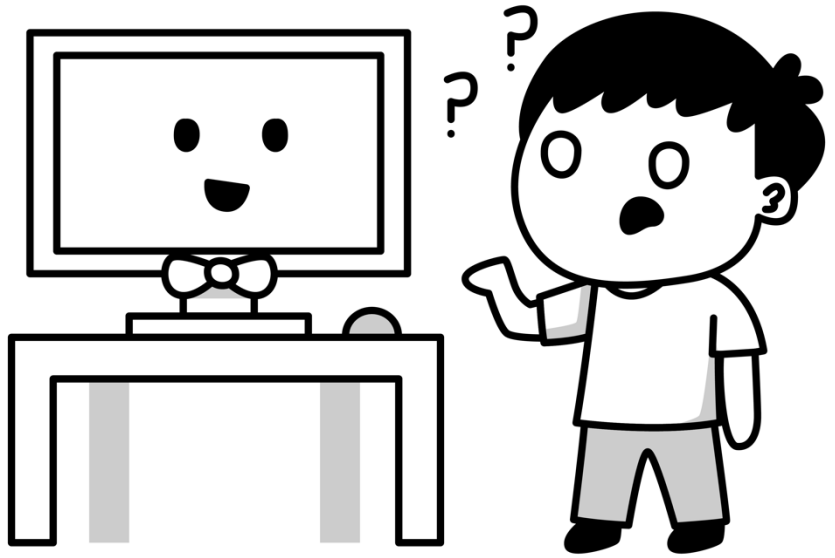
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Demo Time

Please Pray

So what's next?

- DuckDuckGo probably isn't going to move away from Icinga any time soon
- But we can still learn from it!
- onotify will continue to evolve, as an example of what we could have
- Prometheus and its children are *not* the be all and end all



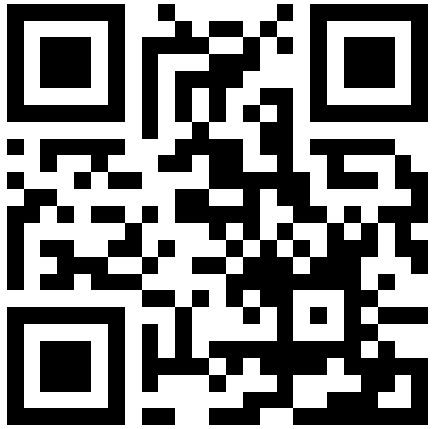
Questions?



https://github.com/sinkingpoint/onotify



Art by Alicia Edwards:
https://www.aliciaillustration.com/



https://colindou.ch/slides

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